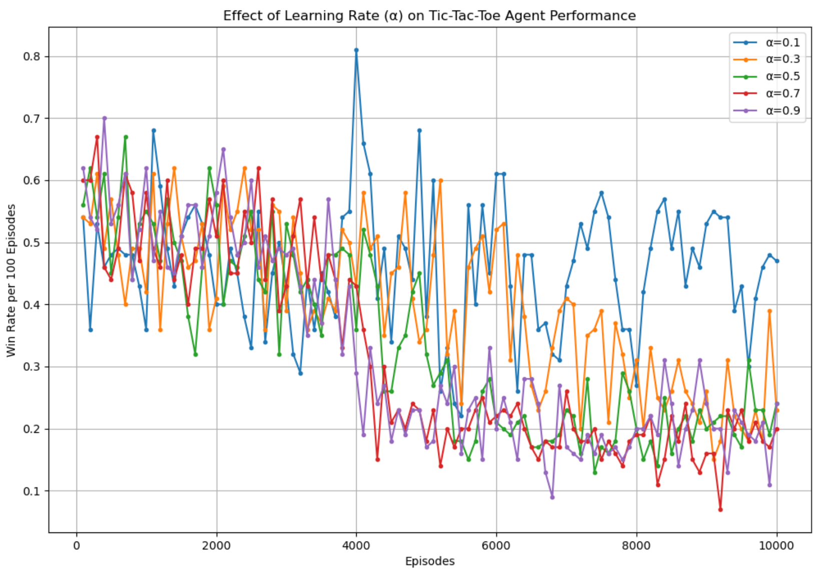
Observe the effect of learning rate, α, and illustrate the outcome of different learning rate with a plot of the winning rate. Specifically, plot the winning rate per 100 episodes over number of training episodes at every 100 episodes ([100, 200, 300, ….])



1. Overall Relationship:

   - A clear inverse relationship exists between learning rate (α) and winning rates
   - As α increases, the winning rates consistently decrease
   - Lower α values (0.1-0.3) achieve significantly better performance than higher values (0.7-0.9)

2. Detailed Observations:
   - α = 0.1: Achieves highest winning rates
   - α = 0.3: Shows slightly lower but still strong performance
   - α = 0.5: Demonstrates moderate performance
   - α = 0.7: Shows declined performance
   - α = 0.9: Results in lowest winning rates

3. Performance Patterns:

   - Early Episodes (1-1000):
     - All α values show rapid initial improvement
     - Higher α shows steeper initial learning curve
   - Mid Episodes (1000-4000):
     - Lower α continues steady improvement
     - Higher α shows performance degradation
   - Late Episodes (5000+):
     - Lower α maintains stable, higher winning rates
     - Higher α settles at lower winning rates with more volatility